

Part 01

The National Bank Group has asked you to build the data layer for their new Bank Management System. Branches across the country need a single, reliable system that tracks branches, managers, customers (individuals and businesses), bank accounts (including joint accounts shared by multiple customers), and every transaction that moves money in or out of those accounts.

Your job is not to build the UI or the ASP.NET API — your job is to build the EF Core Code-First data layer that a future team will plug into. That means:

1. Translating the provided ER diagram into C# entity classes.
2. Configuring the relationships correctly using the Fluent API (and/or Data Annotations).
3. Using Migrations to generate the database.
4. Seeding realistic test data.

Resources :

1. An ER diagram (schema) — the source of truth for structure.
2. A case study document — the source of truth for business rules.

Part 02

After Finishing Database Creation

You must seed at least two tables [Branch - Manager] with initial data

Part 03

Functional Operations via Interactive Menu :

Required menu:

Behavior requirements:

#	Menu Option	What the program must do
1	Add a new Customer	Prompt the user for full name, national ID, DOB, email, phone, address, and customer type (Individual / Business). Validate inputs.
2	Open a new Account for a Customer	Prompt for account number, account type, branch code, the customer Id, and ownership role (Primary / CoHolder). Must verify branch and customer exist before creating the Account
3	Update Account Status	Prompt for account number + customer Id, then toggle AccountStatus
4	Remove an Account from a Customer	Prompt for account number + customer Id, delete the CustomerAccount
5	List all Customers (with accounts)	load each Customer's CustomerAccount rows and the Account behind each. Print a tidy formatted list.
0	Exit	Break out of the loop and close the Connection

Menu-loop quality requirements :

1. Invalid input (e.g., letters where a number is expected) must not crash the app — show a friendly error and re-prompt.
2. After each operation, pause with "Press any key to return to the menu..." before redrawing the menu, so output is readable.

Part 04

Sample Expected Outputs :

Initial screen:

```
=====
National Bank - Management
=====
1) Add a new Customer
2) Open a new Account for a Customer
3) Update Account Status (Active / Closed)
4) Remove an Account from a Customer
5) List all Customers (with accounts)
0) Exit
-----
Enter choice:
```

Option 1 — Add Customer:

```
--- Add New Customer ---
Full Name       : Mona Ahmed
National ID     : 14785236987
Date of Birth   : (yyyy-MM-dd) 2000-01-25
Email           : Mona@gmail.com
Phone           : 0111234567
Address         : 123-Street-City-Country
Customer Type:
    1) Individual
    2) Business
Choice: 1

Customer created successfully. CustomerId = 3

Press any key to return to the menu...
```

Option 2 — Open Account:

```
--- Open New Account ---  
Account Number : 123456  
Account Type:  
    1) Savings  
    2) Current  
    3) Business  
Choice: 1  
Branch Code    : CAI-01  
Customer Id    : 3  
Ownership Role:  
    1) Primary  
    2) CoHolder  
Choice: 1  
Validating branch 'CAI-01' and customer #3...  
Account '123456' created and linked to customer 3 as Primary owner.  
  
Press any key to return to the menu...
```

Option 3 — Update Account Status:

```
--- Update Account Status ---  
Account Number : 123456  
Customer Id    : 3  
New Status:  
    1) Active  
    2) Closed  
Choice: 2  
Status updated to Closed.  
  
Press any key to return to the menu...
```

Option 4 — Remove an Account from a Customer :

```
--- Remove Account From Customer ---  
Account Number : 123456  
Customer Id    : 3  
  
Ownership link deleted.  
That was the last owner - account '123456' was also removed.  
  
Press any key to return to the menu...
```

Option 5 — List customers:

```
--- All Customers ---  
  
#1 Ahmed Ali (Individual)  
   2001-CUR   Current   Balance:   15,400.00   Primary   Active   @ Cairo Main Branch  
#2 Nile Trading LLC (Business)  
   3000-BUS   Business   Balance:  120,000.00   Primary   Active   @ Alexandria Branch  
#3 Mona Ahmed (Individual)  
   (no accounts)  
  
Press any key to return to the menu...
```

ROUTE